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Human appropriation of net primary production in Bangladesh, 1700–2100

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ABSTRACT

In the global debate on environmental integrity, land system is one of the nine planetary boundaries. Bangladesh being one of the top densely populated but rapidly developing countries, lacking in statistical inventories, has a gloomy history of agricultural and infrastructural intensification at the expense of forest cover loss. However, to what extent its land has lost productivity due to land use changes is yet to be calculated for Bangladesh which this study explores for 400 years (1700–2100) through the lens of human appropriation of net primary productivity (HANPP). It shows an increase in both spatial HANPP from 1700 (275.94 gC/m^2) to 1992 (594.38 gC/m^2) and non-spatial HANPP from 1961 (1169.37 gC/m^2) to 2016 (1318.14 gC/m^2) based on which future projections have been made. Population is a reliable predictor of HANPP which indicated the expected HANPP of 1458.20 gC/m^2 by 2100. Bangladesh shows much higher HANPP compared to its peers and hence the approaches and methodologies of this research may help countries with poor statistical database to calculate HANPP. This study rendered where and to what degree the net primary productivity was appropriated in Bangladesh, suggesting region specific policy and programme interventions necessary to reduce HANPP. The GDP-HANPP regression model indicates a loss of approximately 8.62 million 2018 US$ GDP for reduction of HANPP by 1 kt C/year which can be used as a guiding value in seeking fund from carbon financing mechanisms to pursue the gross HANPP reduction threshold of 926.675 gC/m^2/year corresponding to the achievement of 25% forest cover in Bangladesh. The HANPP case of Bangladesh provides clue to countries witnessing rapid population and economic growth and associated with large-scale land use changes to expand agriculture and infrastructure, indicating that inaction on their part in harmonizing development planning based on HANPP indicators may jeopardize the attainment of SDGs.

1. Introduction

One of the largest contributors to global environmental change in this “age of the Anthropocene”, is land-system change. Accordingly, it is considered as one of the nine planetary boundaries that if crossed can render the planet inhabitable for humans (Steffen et al., 2015). Conversion of forest ecosystem for agricultural expansion and intensification is leading to loss of ecosystem functioning, decreasing the integrity of the biosphere and impeding the regulatory apparatus of the Earth by impacting its climate system and biogeochemical cycles (Haberl et al., 2014; Steffen et al., 2015). High productivity land under forest ecosystems are increasingly being converted for anthropogenic purposes, such as food production, urbanization, fuel generation, etc., leaving lesser biomass for all other species of the planet (Foley et al., 2005). In that sense, humans are appropriating a non-substitutable biologically productive land resource that is crucial for the functioning not only of socioeconomic but also of ecological systems encompassing all terrestrial biomes. Hence, the spatial distribution and intensity of land-system change needs to be regulated which in turn will help regulate the distribution of Earth’s biological capacity, across all species and ecosystems (Haberl et al., 2014).

In order to hold accountable, the ever-increasing human appropriation of the earth’s ecosystem, be it in terms of natural resources, capital, or energy flows, a family of footprint indicators such as ecological footprint, water footprint, carbon footprint, material footprint, human appropriation of renewable freshwater, land footprint of human activities, indicators of actual land demand and many others (Haberl et al., 2014) have evolved over the last few decades, each focusing a specific environmental concern, resource or approach of quantification (Ridoutt and Pfister, 2013). Among them, few indicators, mainly

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ecological footprint and human appropriation of net primary productivity (HANPP), which specifically focus on primary production and relate it with socio-economic metabolism are crucial since land use change and associated ecosystem transitions depend on socioeconomic factors (Lambin and Meyfroidt, 2010). However, these two indicators investigate different types of ecological impacts and in turn provide different insights into sustainability. Unlike ecological footprint that measures the area of land required to maintain the socioeconomic metabolism of a given population, HANPP measures the intensity of impact of that population on that specific area of land. It is then compared with the total potential biological productivity of an ecosystem which would be available, in the absence of humans. Therefore, HANPP as an indicator provides powerful insights of the level of human domination in terms of the size of anthroposphere within the earth's ecosystem. An increase in HANPP means a decrease in the flow of materials, substance and energy for non-human processes resulting in lower sustainability potential of the overall ecosystem.

Even though, analysis of HANPP is not very straightforward, there are many practical implications that have been utilized by countries to redesign their policies for more efficient land use to effectively reduce HANPP (Niedertscheider et al., 2012; Schwarzlmüller, 2009; Kastner, 2009; Kohlheb and Krausmann, 2009). The efficacy of HANPP as an indicator lies both in the reality check that it provides and the questions it answers regarding the efficiency of specific or general land use.

Global enhancement of forest land transformation into agricultural land is evident fromincreasing HANPP, which was the highest in the South Asian region (Haberl et al., 2007). This observation has later been confirmed by country level studies in Philippines (Kastner, 2009) and in India (DeSouza and Malhi, 2018). It will be interesting to see the HANPP for Bangladesh - the most densely populated country in South Asia (UN, 2017) which is small in terms of land area yet getting global attention as one of the emerging economies of the world due to sustained GDP growth over 6% since 2011 (World Bank, 2017) - as the country's fertile terrestrial biomes having significant primary productivity are getting transformed at unprecedented rate for food production and to meet infrastructural needs of the growing population of over 163 million (UN, 2017) with increasing living standards. One of the first studies on land use change in Bangladesh (Richards and Flint, 1994) showed a visible declining trend in the total forest cover of the country with corresponding increase in the total cropland. Considering the sharpest increase in GDP growth rate over the last decade that has created the expectation of becoming a Lower Middle Income Country (LMIC) by 2024 (United Nations, 2019). Consequently, this research focused into measuring the impacts of this stunning growth by using HANPP as an indicator to help refine the global estimates by providing country specific data from this region while gaining relevant policy insights that may help contextualize the country's development planning without jeopardizing the attainment of sustainable development goals linked to the sustenance or enhancement of net primary productivity. The implications drawn out from the results of this research may also contribute to the global discussion needed to contain the land-system boundary within permissible limits.

2. Materials and methods

The synopsis of the approach of the study has been illustrated as a process flow and can be found in supporting information Figure S1.

2.1. General overview

This study utilizes the definition of HANPP provided by Haberl et al. (2014) and the approach to calculate it by DeSouza and Malhi (2018) in order to assess the scenario of Bangladesh. HANPP is described as the difference between the net primary productivity (NPP) of pristine floras in the absence of human induced land use and the NPP that actually exists in the land system after harvest under present conditions (Haberl et al., 2014). The value of HANPP can be derived as follows:

$$HANPP_{luc} = NPP_{potential} - NPP_{actual} \quad (1)$$

$$HANPP_{harvest} = NPP_{actual} - NPP_{eco} \quad (2)$$

$$HANPP = HANPP_{luc} + HANPP_{harvest} \quad (3)$$

$$HANPP = NPP_{potential} - NPP_{eco} \quad (4)$$

$NPP_{potential}$ is the NPP of natural vegetation, the amount of biomass that would have been stored in vegetation in the absence of humans and NPP_{actual} is the amount of NPP that remains in the ecosystem after anthropogenic intervention (e.g., cultivation or clearing of forest). $HANPP_{luc}$ is the part of HANPP which defines the change of NPP occurred due to any sort of land use conversion. NPP_{eco} is the component of NPP_{actual} which remains after harvest (e.g., vegetation in agroecosystem, non-timber trees in forest harvesting area) and $HANPP_{harvest}$ is the element of HANPP related to the harvest biomass including crop residues (DeSouza and Malhi, 2018) (to see the comparison of the different components of HANPP calculations with the total potential NPP, please refer to supporting information Figure S2).

2.2. Data

For a better understanding about the complex dynamic relationship between human dimension and the determinant factors of land use alteration, detailed geographic information is required for analyzing the anthropogenic interferences which influence global ecological processes (Erb et al., 2007). Therefore, we utilized two approaches viz., i) using *spatially explicit data*[1] and ii) using *non-spatially explicit data*[2] for calculations over a historical period of time (1700–2016). Since land use data from local sources were mostly unavailable and tend to be erroneous, especially in case of developing countries like Bangladesh, we opted to use global scientific dataset instead. For the quantification of HANPP, the components are calculated with the help of various datasets. A comprehensive explanation of the calculations and the data sources has been summarized in Table 1.

2.3. Calculating NPP of Bangladesh

2.3.1. Potential net primary productivity ($NPP_{potential}$)

Malhi and colleagues (2015, 2016) show that in tropical regions NPP is 40% of gross primary productivity (GPP). Hence to calculate the potential GPP ($GPP_{potential}$) of Bangladesh we followed the model of a study carried out in India (DeSouza and Malhi, 2018) since both the countries are in the same tropical region and because it overcame uncertainties existing in previous models. However, in terms of dataset we used the more updated version of global GPP (gC/m^2/year) dataset (D1, Table 1) at 0.5 min resolution derived from MODIS satellite from Zhang et al. (2017), while precipitation dataset (D2, Table 1) was derived from Fick and Hijmans (2017). The GPP of each pixel within Bangladesh was plotted against corresponding mean precipitation. $GPP_{potential}$ was calculated from the upper envelope of the plot, assuming that it was mostly influenced by precipitation (for details, see supporting information Figure S3). Finally, the $NPP_{potential}$ of Bangladesh was obtained multiplying $GPP_{potential}$ by the carbon efficiency.

2.3.2. Actual net primary productivity (NPP_{actual})

NPP_{actual} was calculated using the formula mentioned by DeSouza and Malhi (2018):

$$NPP_{actual} = \sum_c NPP_c \times Area_c \quad (5)$$

[1] Using gridded data with spatial coordinates of global cropland dataset from Ramankutty and Foley (1999)

[2] Using non-gridded data of total land system of Bangladesh from FAO (2017)

Table 1
Details of datasets.

Components of HANPP	Landuse type NPP calculation	Influencing parameters	Dataset(D)		Details of calculation
Potential NPP	$NPP_{potential}$ (gC/m^2/year)	GPP (gC/m^2/year)	Global GPP (gC/m^2/year) dataset at 0.5 min resolution	D1	Precipitation vs GPP was plotted. $GPP_{potential}$ was obtained from the upper envelope of this graph.
		Precipitation (mm)	Precipitation (mm) at 30 s resolution	D2	Malhi and collegeues (2015, 2016) showed that the NPP can be found from GPP using carbon efficiency and for tropical forest it is 40% or 0.4 (DeSouza and Malhi, 2018)
Spatial actual NPP	Spatial Cropland NPP	Crop land gridded area (m^2)	Ramankutty and Foley dataset	D3	Area multiplied by yield (incorporating aboveground and belowground biomass) gives the cropland NPP (DeSouza and Malhi, 2018)
		Cropland yield (g/m^2)	Major crops production and area dataset of Bangladesh	D4	
		C3 and C4 Crop NPP	ORNL DAAC Net Primary Production data	D5	
		C3 and C4 crop distribution	Global distribution of C3 and C4 vegetation	D6	
	Spatial Forestland area	Bangladesh forestland gridded area (m^2)	Ramankutty and Foley dataset	D3	NPP of forestland is calculated by multiplying the area (D3) of natural vegetation by the corresponding value of $NPP_{potential}$ for the grid cell (DeSouza and Malhi, 2018)
		Forestland NPP (gC/m^2/year)	Potential NPP of Bangladesh	Derived from D1 and D2	
Non-spatial actual NPP	Cropland NPP	Cropland area (m^2)	FAO dataset	D7	Area times yield gives the NPP of cropland
		Cropland yield (g/m^2)	FAO dataset		
	Pastureland NPP	Pastureland area (m^2)	FAO dataset	D7	Assuming pastures are C3 plants (DeSouza and Malhi, 2018), we calculate the NPP of pasturelands as area times average total NPP of C3 grasslands.
		C3 grasslands total average NPP (g/m^2)	ORNL DAAC Net Primary Production (NPP) data	D5	
	Forestland NPP	Forestland area (m^2)	Bangladesh forestland area dataset	D8	Since no study was done on the NPP of forests of Bangladesh, we use the potential NPP value of the forest region following Haberl et al. (2004 and 2005).
		Forestland NPP (gC/m^2)	Mean NPP of forestland	Derived from D1 and D2	
	NPP appropriated by harvest biomass	Harvest index (HI)	Krausmann et al. (2008) harvest index list	D9	To calculate the harvest biomass NPP, we calculate the entire dry matter yield through Lobell et al. (2002) equation
		Mass per each component of given yield for each crop (MRY) (g/m^2)	FAO dataset	D7	
		Moisture Content (MC)	Lobell et al. (2002) MC list and USDA Food water content dataset	D10	
	NPP appropriated by livestock demand	Feed conversion factor (g/head)	Feed conversion factor for South Asia region	D11	The quantity of NPP appropriated by livestock can be obtained by livestock population times feed conversion factor. Part of this demand provided from crop residues, market feed and non-market feed. Here there is no official dataset for nonmarket feed for Bangladesh.
		Livestock population	Rahman et al. (2014)	D14	
			FAO dataset	D7	
		Market feed (g/m^2)	FAO dataset	D7	
		Non market feed (g/m^2)	No official data		
Embodied HANPP		Crops imports NPP (gC/m^2/year)	FAO dataset	D7	Erb et al. (2009) definition was used to calculate the embodied HANPP
		Crops exports NPP (gC/m^2/year)			
Future predictions of spatially explicit HANP		Spatial HANPP	Spatial HANPP (1700-1992) calculated prior to future projection	Derived from D3, D4, D5 and D6	The hypotheses function provides future prediction of HANPP given input x and weight vector θ, which is derived using the normal equation.
Future predictions (embodied and non-spatial HANPP)	Population	Bangladesh population dataset (2016- 2100)	Population Estimates and Projections	D12	Considering Krausmann et al. (2013), the predicted values are extrapolated against four independent variables i.e. time, population, GDP, biomass consumption (Chow and Lin, 1971)
	GDP (US$)	Bangladesh GDP dataset (2016- 2100)	socio-economic projections	D13	
	Biomass consumption (g/capita/year)		FAO dataset	D7	

(*continued on next page*)

Table 1 (*continued*)

Components of HANPP	Years calculated	Equation (number)		Reference
Potential NPP	N/A	$Mean\ GPP_{potential}\ of\ Bangladesh = \frac{\sum(GPPpotential \times Area\ of\ the\ respective\ grid\ cell)}{total\ area\ of\ Bangladesh}$	6	Zhang et al. (2017)
	N/A	$NPP_{potential} = 40\%\ of\ Potential\ Gross\ Primary\ Productivity$	7	Fick and Hijmans (2017)
Spatial actual NPP	1700-1992	$Total\ NPP\ of\ cropland = Area \times yield \times \frac{Total\ NPP\ of\ crops}{Aboveground\ NPP\ of\ crops}$	8	Ramankutty and Foley (1999)
	1700-1992			Kurosaki (2011)
	1700-1992			ORNL DAAC (2019)
	1700-1992			Still et al. (2003)
	1700-1992	$NPP_{forestland} = Area\ of\ the\ grid\ forestland\ area \times NPP_{potential}\ of\ the\ corresponding\ grid$	9	Ramankutty and Foley (1999)
	1700-1992			
Non-spatial actual NPP	1961-2016	$NPP_{cropland} = Area \times yield$	10	FAO (2017)
	1961-2016	$NPP_{pastureland} = 1300gm^{-2} \times Area$	11	FAO (2017)
				ORNL DAAC (2019)
	1961-2016	$NPP_{forestland} = Area \times NPP_{forestland}$	12	Reddy et al. (2016)
	1961-2016	$NPP_{harvest} = \frac{MRY \times (1 - MC)}{HI}$	13	Krausmann et al. (2008)
				FAO (2017)
				Lobell et al. (2002)
				USDA (2018)
	1961-2016	$NPP_{demand} = (Feed\ conversion\ factor \times Livestock\ population) - Crop\ residue\ derived\ feed - market\ feed - non\ market\ feed$	14	Krausmann et al. (2008)
				Rahman et al. (2014)
				FAO (2017)
Embodied HANPP	1961-2013	Embodied HANPP = HANPP + NPPimports – NPPexports (Eq. 13)	15	FAO (2017)
Future predictions of spatially explicit HANP	1993-2100	$h_\theta(x) = \theta_0 + \theta_1 x$	16	Kenney and Keeping (1962)
		$\theta = (X^T X)^{-1} X^T Y$	17	
Future predictions (embodied and non-spatial HANPP)	2017-2100 (non-spatial) 2014-2100 (embodied)	$\boldsymbol{y}(\boldsymbol{x}) = \boldsymbol{y}_1 + \frac{\boldsymbol{x} - \boldsymbol{x}_1}{\boldsymbol{x}_2 - \boldsymbol{x}_1}(\boldsymbol{y}_2 - \boldsymbol{y}_1)$	18	World Bank (2017)
				Graffin et al. (2004)
				Geiger (2018)
				FAO (2017)
				World Bank (2017)
				Gaffin et al. (2004)

Here, c represents each type of land use classification and $Area_c$ indicates the land area of that particular land use type. In the spatial approach (spatial NPP_{actual}), land use was classified as (i) cropland and (ii) forestland whereas in the non-spatial approach (non-spatial NPP_{actual}), land use was classified as (i) cropland, (ii) forestland, (iii) pastureland and (iv) other lands.

To calculate spatial NPP_{actual}, the Ramankutty and Foley (1999) dataset (D3, Table 1) was used, which is a global comprehensive dataset of croplands from 1700 to 1992. Any remaining land was considered as forestland (gridded map of Bangladesh was clipped from the dataset and area which was not cropland was considered as natural vegetation), including the urban or built environment which encompasses a small portion of Bangladesh, approximately 6% (Bangladesh Bureau of Statistics, 2012). To calculate spatial NPP_{actual} of cropland, the values of area, yield as well as above and below ground biomass are required (Eq. 5, for further details see Table 1). There is no gridded data on crop yield for Bangladesh and hence an annual average of the yield for the total land cover of Bangladesh was calculated for the years 1700-1992. The yield data (in g/m^2) was estimated by calculating production of crop per unit area, the data for which was obtained from Kurosaki (2011) (D4, Table 1). However, Kurosaki (2011) provides the data of area and crop production of Bangladesh from the year 1901 to 2001. Therefore, for the years 1700 to 1900 we considered the average crop yield value of 1901 as the constant. It should be noted that the product of area and yield gives only above-ground NPP, excluding non-useful parts of the vegetation from the calculation and thus the ratio of total NPP of crops and above-ground NPP of crops was calculated using DeSouza and Malhi (2018)'s model (for the equation and other details please see Table 1). Oak Ridge National Laboratory (ORNL DAAC, 2019) provided the NPP estimation of C3/C4 plants (D5, Table 1) and Still et al. (2003) mapped those plants globally using Ramankutty and Foley (1999) dataset (D3, Table 1). Their map (D6, Table 1) shows the C3/C4 ratio of Bangladesh showing C3 plants to mostly dominate with a small proportion of C4. The ratio of total to above-ground NPP of Bangladesh is assumed as 3 following the study of HANPP in India carried out by DeSouza and Malhi (2018). The spatial NPP_{actual} of forestland is calculated by multiplying the area (D3, Table 1) of natural vegetation by the corresponding value of $NPP_{potential}$ for each grid cell (equation 9, Table 1). To calculate non-spatial NPP_{actual}, we relied mostly on FAO dataset (D7, Table 1). Non-spatial NPP_{actual} of cropland is calculated by multiplying area with yield (equation 10, Table 1). Here, cropland area is defined as the sum of the arable land and permanent crop area and mean yield was calculated by dividing the total crop production by total area of cropland. To get the value of non-spatial NPP_{actual} of forestland, we needed values of forest area and the mean NPP value of forestland (equation 12, Table 1). Since forestland area of Bangladesh before 1990 was unavailable in FAO dataset, we used Reddy et al. (2016) dataset (D8, Table 1) for the calculation of non-spatial NPP_{actual} of forestland. There has been no previous study on forestland NPP of Bangladesh, therefore assuming that the values of $NPP_{potential}$ and NPP derived from forest inventory data are almost similar (as seen in Haberl et al., 2004, 2005), we used the dataset from $NPP_{potential}$ to calculate non-spatial mean NPP of forestland (which was 622.5 gC/m^2 for Bangladesh and we found it consistent with that of the value of India's data which is $591 gC/m^2$ (Prasad and Badarinth, 2004)).

In case of non-spatial NPP_{actual} of pastureland, the area was multipliedwith NPP of pasture crops (equation 11, Table 1). We assumed that pastures are C3 plants (DeSouza and Malhi, 2018), so the area (D7, Table 1), derived from FAO dataset, was multiplied by $1300 g/m^2$ which is the average total NPP (D5, Table 1) of C3 grasslands (ORNL DAAC, 2019).

For the calculation of non-spatial NPP of harvest biomass ($NPP_{harvest}$), we used the equation from Lobell et al. (2002) (equation 13, Table 1) for which the mass per unit of yield of each crop type (D7, Table 1) (FAO, 2017), moisture content (D10, Table 1) (USDA, 2018; Lobell et al., 2002) and the harvest index (D9, Table 1) (Krausmann et al., 2008) were required. Harvest index is the economic yield of the crop expressed as a fraction of the total biological yield which involves the entire dry mass a vegetation produces above ground. Krausmann et al. (2008) provides a detailed list of harvest factors (D9) for several crops for the South Asian region.

The net primary productivity appropriated by livestock is defined as NPP_{demand} following the model of Krausmann et al (2008) and it can be calculated by multiplying livestock feed demand (D11, Table 1) (data of livestock feed demand for the South Asian region was obtained from Krausmann et al (2008)) and the population of livestock (D14: 1961–1979 from Rahman et al., 2014 and D7: 1979–2016 from FAO, 2017) (equation 14, Table 1). This paper puts forward, that a portion of this demand is provided by sources other than pasturelands such as crop residues, market feed and non-market feed which need to be factored out of the calculation to avoid overestimation. From DeSouza and Malhi (2018) we found that 83% of the crop residues are used as livestock feed. We found the data on market feed (D7, Table 1) for Bangladesh from the year 1961 from FAO database but there was no official estimate of non-market feed thus it was ignored in this study. However, it should be noted that majority of the livestock of Bangladesh are raised in grassroots small-scale farming families and so their agricultural calculations are not included in the FAO dataset. Thus, while calculating the NPP_{demand} appropriated by livestock, the data of small-scale farming croplands possibly gets included (DeSouza and Malhi, 2018). Nevertheless, a detailed calculation of non-market feed of Bangladesh would refine this dataset. The NPP of other lands (urban/built areas) were considered zero since the urban area of Bangladesh is very low (about 6%), as mentioned earlier. This assumption of taking urban land NPP as zero does not portray the true picture since there is presence of greeneries within the urban boundary such as parks, gardens, rooftop gardens etc. But due to lack of data it had to be excluded from our calculations.

2.4. Calculating human appropriation of net primary productivity

Spatially explicit HANPP was calculated based on the area dataset of Ramankutty and Foley (1999) using Eq. 1. Note that this definition of HANPP does not incorporate harvest biomass and livestock demand, which we overcame by including non-spatially explicit HANPP (using the Eq. 4 predominately through FAO dataset). The NPP appropriated from forest was not incorporated in this definition due to unavailability of data. However, the demand of this country's population on the ecosystem of other territories (imports) cannot be included and the demand on this country by people from outside territories (exports) cannot be excluded in the conventional HANPP approach. That is where embodied HANPP (Eq. 15, Table 1) comes of use since it incorporates the entire biomass removal and land use induced alterations in productivity occurring from the use of biomass commodities in Bangladesh (Erb et al., 2009). Exports and imports of crop products from and to Bangladesh were collected from FAO dataset (D7). The approach used in calculation of NPP of harvest biomass was used to calculate the NPP of the imports and exports.

2.5. Future projections

2.5.1. Spatial HANPP

The spatial HANPP of Bangladesh is derived as raster data from Ramankutty and Foley (1999). Each year's raster data is in the form of 5,519 georeferenced pixels with corresponding HANPP values.

An Ordinary Least Squares (OLS) regression model has been used for the modelling and future prediction of spatial HANPP of Bangladesh. Each pixel location is modelled using an OLS regression model, resulting in the training and testing of 5,519 separate models, each provided with HANPP values of the corresponding pixel in the temporal domain of 292 years (1700–1992) (D3). A closed-form solution of OLS regression (Kenney and Keeping, 1962) was applied. Due to use of

single predictor variable, feature normalization is unnecessary (further unnecessary due to use of normal equations in obtaining weights) and a univariate linear regression model is generated. The hypotheses function, $h_\theta(x)$, with intercept (bias) term θ_0, and a gradient (coefficient) term θ_1 is shown in equation 16 (Table 1).

In this case, x represents the input (time in years) and $h_\theta(x)$ represents predicted HANPP values. The solution via normal equations which involves finding values of θ_0 and θ_1, can be obtained using equation 17 (Table 1), where X is the input matrix (with added column of 1's for bias term) and Y is the response vector (HANPP) and θ is the vector of weights. The 5,519 models are trained and tested using HANPP data spanning 292 years.

The provided data samples are split using 80:20 ratio for training and testing respectively. The input feature i.e. the independent variable can be considered as time (in years) and the output/target/response i.e. dependent variable is the HANPP value for the corresponding pixel location. The trained and tested model has been used for predicting the HANPP values of each pixel location for the years 2000, 2013, 2015, 2016, 2018, 2019, 2020, 2030, 2040, 2050, 2060, 2070, 2080, 2090 and 2100. The predicted values are used for creating raster maps showing the future trends and distribution of HANPP of Bangladesh.

2.5.2. Non-spatial HANPP

Three independent variables (see Table 2) have been considered for the purpose of predicting future values of non-spatial and embodied HANPP viz., i) Gross Domestic Product (GDP) (D13), ii) population (D12) and iii) biomass consumption per year (D7). Such features were also employed in Krausmann et al., 2013 due to having strong relationship with HANPP.

Linear extrapolation (Chow and Lin, 1971) has been used shown in equation 18 (Table 1). Considering $y(x)$ to be the predicted HANPP value corresponding to input x. x_1 and y_1 represents a temporal predecessor input and corresponding HANPP value respectively. x_2 and y_2 are analogous to x_1 and y_1, but having the difference such that $x_2 > x_1$.

Univariate linear extrapolation has been used for the modelling and prediction of non-spatial and embodied HANPP which have been trained using data from a temporal domain of 52 years (1961 to 2013). The non-spatial HANPP of Bangladesh is predicted for the years 2018, 2019, 2020, 2030, 2040, 2050, 2060, 2070, 2080, 2090 and 2100 while the embodied HANPP is predicted for the years 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2030, 2040, 2050, 2060, 2070, 2080, 2090 and 2100.

2.6. Scenario based predictions

A scenario based analysis of non-spatial and embodied HANPP is done by exploring different factors that may affect HANPP individually or altogether such as GDP, population, biomass consumption. Scenarios are explored by extrapolating against each of the independent variables (Krausmann et al., 2013). An aggregated trend is derived from the three different scenarios for generating concrete future predictions.

2.7. Classification based on HANPP and Human Development Index (HDI)

Bangladesh has 64 administrative regions known as districts. In order to assess the implications of HANPP on a local scale i.e. district-based analysis, it is important to classify and differentiate these districts based on socio-economic factor as well as HANPP. The HDI (which is a statistical composite of life expectancy, education and per capita income is chosen since HANPP itself is indirectly linked to aforementioned factors as well as land use change and biogeochemical properties). Similar comparison based on HANPP and HDI was shown in DeSouza and Malhi (2018). The district level HDI data of the year 2000 is combined with the spatial HANPP for the same year (following Khatun, 2001) since HDI scores for all 64 districts are available only for that year. For this analysis four classes/categories are defined viz., i) low HANPP-low HDI (having HANPP range between 0 and 500 gC/m^2/year and HDI range between 0.2 and 0.35), ii) low HANPP-high HDI (having HANPP range between 0 and 500 gC/m^2/year and HDI range between 0.35 and 0.5), iii) high HANPP-low HDI (having HANPP range between 500 and 1000 gC/m^2/year and HDI range between 0.2 and 0.35 and iv) high HANPP-high HDI (having HANPP range between 500 and 1000 gC/m^2/year and HDI range between 0.35 and 0.5).

Table 2
Factors considered for future prediction of non-spatial HANPP.

Scenarios	Factor	Source	Future projection approach
Scenario I	GDP	D13	Linear extrapolation following Krausmann et al. (2013)'s model
Scenario II	Population	D12	
Scenario III	Biomass consumption	D7, D12	
Scenario IV	GDP, population, biomass consumption	D7, D12, D13	Best fit mean of Scenario I, II and III

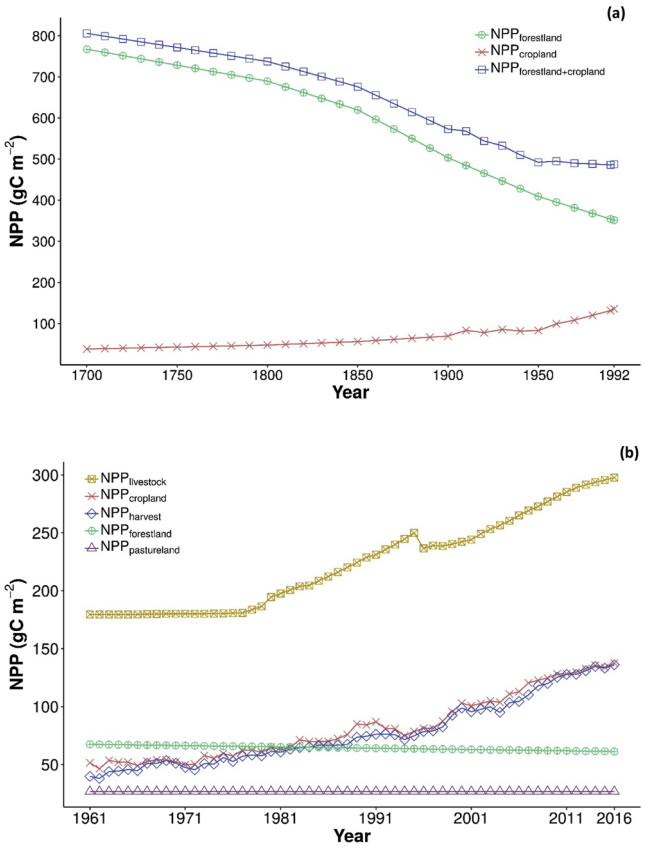


Fig. 1. Components of (a) spatial NPP$_{actual}$ and (b) non-spatial NPP$_{actual}$ of Bangladesh.

2.8. Tools of analysis

The data processing and computation was performed utilizing the scikit-learn package (Pedregosa et al., 2011) of Python 3 language (Van Rossum and Drake, 2003). The geospatial analysis was performed through ArcGIS.

3. Results

3.1. Net primary productivity of Bangladesh

As mentioned in previous section, in order to derive the HANPP of Bangladesh, the calculation of NPP is an integral predecessor step. The study has calculated both the potential and the actual NPP of

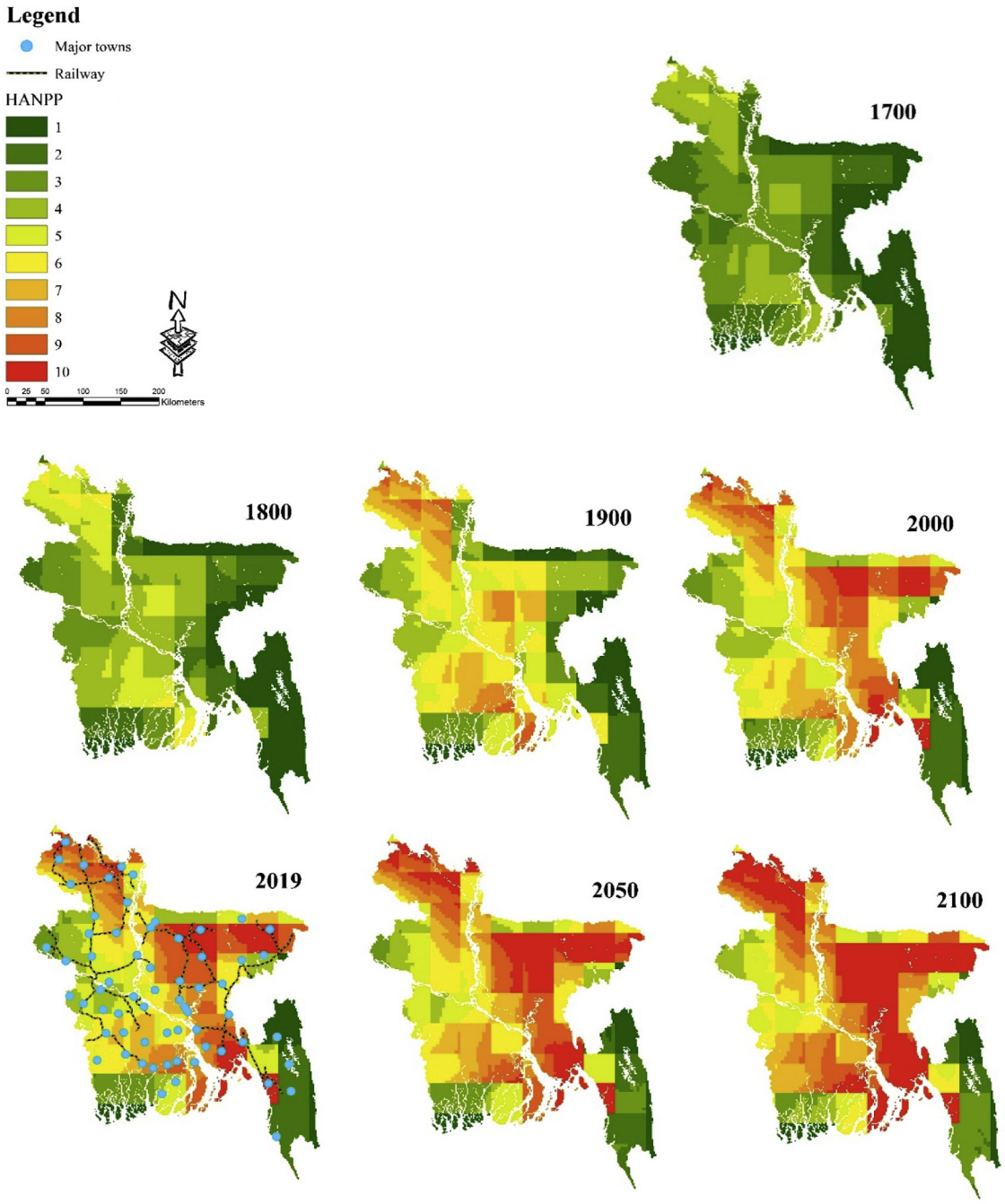


Fig. 2. Calculated spatial HANPP of Bangladesh for 1700, 1800, 1900 and spatial future projections of HANPP for 2000, 2019, 2050 and 2100.

Bangladesh. The $NPP_{potential}$ in turn depends on the $GPP_{potential}$ and the carbon-use efficiency. Our analysis revealed that $GPP_{potential}$ of Bangladesh is 2704.25 gC/m^2/year (for further details see supporting information S3) and therefore, the obtained $NPP_{potential}$ is 1081.7 gC/m^2/year for Bangladesh. This value is in agreement with the $NPP_{potential}$ value of the border region closest to Bangladesh as seen in DeSouza and Malhi (2018) (within the range of 800 and 1200 gC/m^2/year).

We further calculated another component of HANPP, NPP_{actual}, obtained both in spatial and non-spatial context (Fig. 1 shows the components of spatial and non-spatial NPP_{actual}).

The spatial NPP_{actual} (Fig. 1a) was calculated (see Table 1) as the sum of $NPP_{cropland}$ and $NPP_{forestland}$. The comparison of change of spatial NPP_{actual} of cropland and forestland is concurrent with the historical deforestation of Bangladesh (for further details see supporting information Figure: S4). The mean spatial NPP_{actual} of Bangladesh has decreased from 805.75 gC/m^2 in 1700 to 487.21 gC/m^2 in 1992. The components of non-spatial NPP_{actual} of Bangladesh (Fig. 1b) also show a decline in forestland as well as an increase in cropland, concurring with components of spatial NPP_{actual}. In case of Bangladesh, rate of livestock production is the major component of non-spatial NPP_{actual} which is evident from the results in Fig. 1b. Furthermore, the reason behind even higher value of $NPP_{livestock}$ is the absence of nonmarket feed data. Stationarity of $NPP_{pastureland}$ can be attributed to constant pastureland area data in FAO (2017).

3.2. Human appropriation of net primary productivity of Bangladesh

Several datasets have been used for the calculation of HANPP of Bangladesh (mentioned in Table 1). However, irrespective of dataset used or spatial/non-spatial context of calculation, HANPP shows an

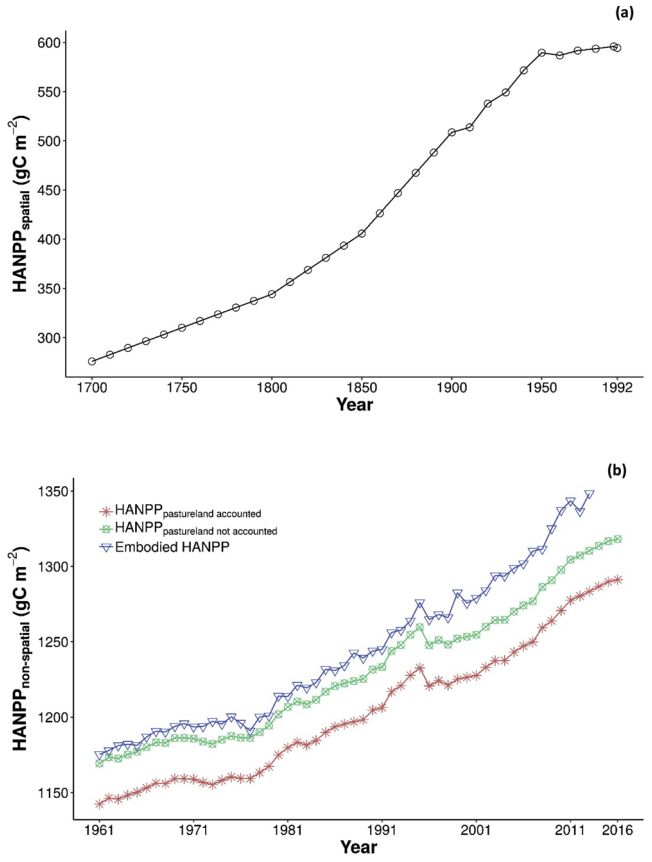


Fig. 3. HANPP of Bangladesh in (a) spatial context from 1700 to 1992 and (b) non-spatial context i.e. HANPP$_{pastureland\ accounted}$ and HANPP$_{pastureland\ not\ accounted}$ from 1961 to 2016 and embodied HANPP from 1961 to 2013.

increasing trend (Figs. 2 and 3).

Figs. 2 and 3a show the increasing pattern of spatial HANPP of Bangladesh from a historical timeline. The value of HANPP has increased drastically from 275.944 gC/m^2/year to 594.38 gC/m^2/year from 1700 to 1992. Fig. 2 also shows the future projections of spatial HANPP which are mentioned in the upcoming subsection. Fig. 3b shows three variants of non-spatial HANPP viz., i) HANPP$_{pastureland\ accounted}$ which accounts for NPP$_{pastureland}$, ii) HANPP$_{pastureland\ not\ accounted}$ which does not account for NPP$_{pastureland}$ and iii) embodied HANPP. HANPP$_{pastureland\ not\ accounted}$ is chosen over HANPP$_{pastureland\ accounted}$ due to constant pastureland area data in FAO (2017). The non-spatial HANPP (with pastureland not accounted) has increased from 1169.370 gC/m^2 in 1961 to 1318.14 gC/m^2 in 2016. Whereas, embodied HANPP has increased from 1175.326 gC/m^2 in 1961 to 1348.45 gC/m^2 in 2013. Noteworthy is that, the spatial HANPP does not incorporate livestock demand and harvest biomass, which explains the fact that spatial HANPP is lower than HANPP calculated using non spatial data for the same time period.

3.3. Socio-economic factors affecting HANPP

We analyzed three factors viz. GDP, population and biomass consumption, which have been linked to HANPP in existing literature (Krausmann et al., 2013), to explore the relationship HANPP exhibits as a function of those socioeconomic factors. Fig. 4 shows the correlation between above said factors and HANPP. High positive value of R^2 for GDP and population, and moderate values for biomass consumption indicate that GDP and population have a determinant role in affecting the HANPP of Bangladesh whereas biomass consumption has a lesser effect.

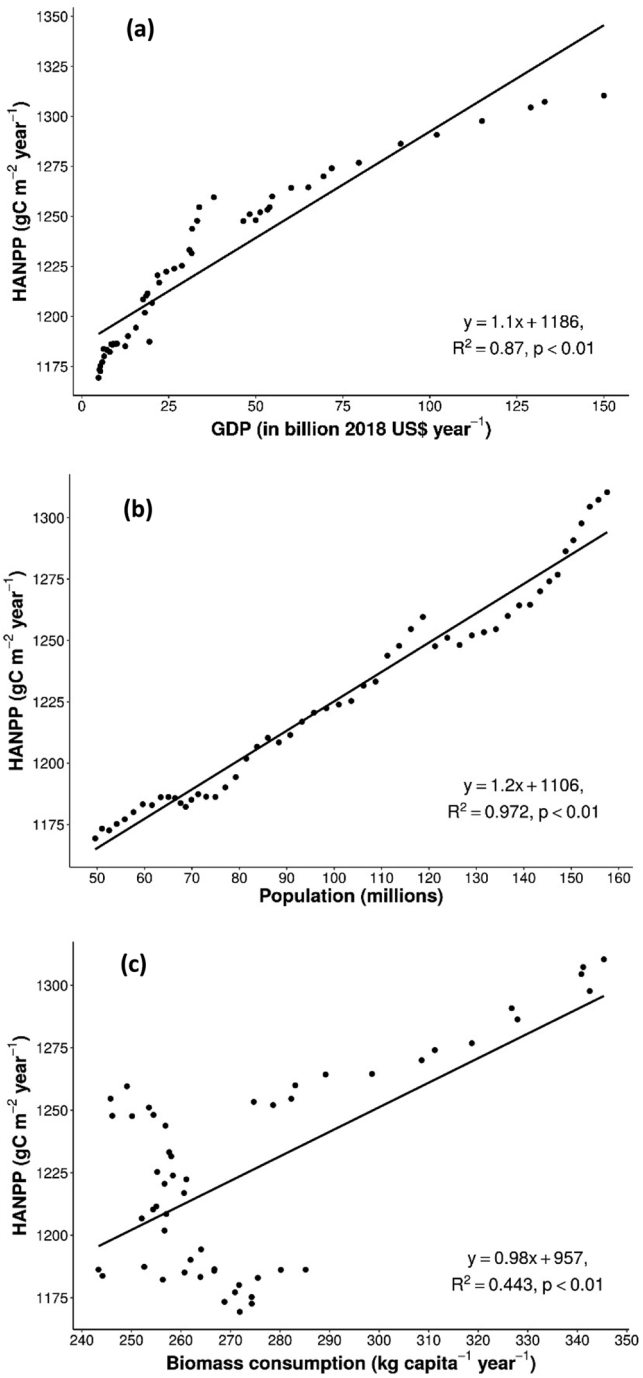


Fig. 4. Relationship between HANPP and socio-economic factors (a) GDP, (b) population and (c) biomass consumption for the temporal interval 1961–2013.

3.4. Future projections of HANPP in Bangladesh

The spatial future projections of HANPP of Bangladesh (Fig. 2) show that the northern, north-eastern, southern and regions surrounding the capital of Dhaka especially the urban and industrial zones are most susceptible of exhibiting high HANPP in the future. The future projections of non-spatial HANPP of Bangladesh based on specific scenarios (mentioned in Table 2) are shown in Fig. 5. Regarding both non-spatial and embodied HANPP, Scenario I (based on GDP) shows exponential growth of HANPP, the same holds true for Scenario II (based on population) but with a lower magnitude than Scenario I. Scenario III (based on Biomass consumption) shows restricted increase in HANPP values in the future which can be linked to the moderate dependence

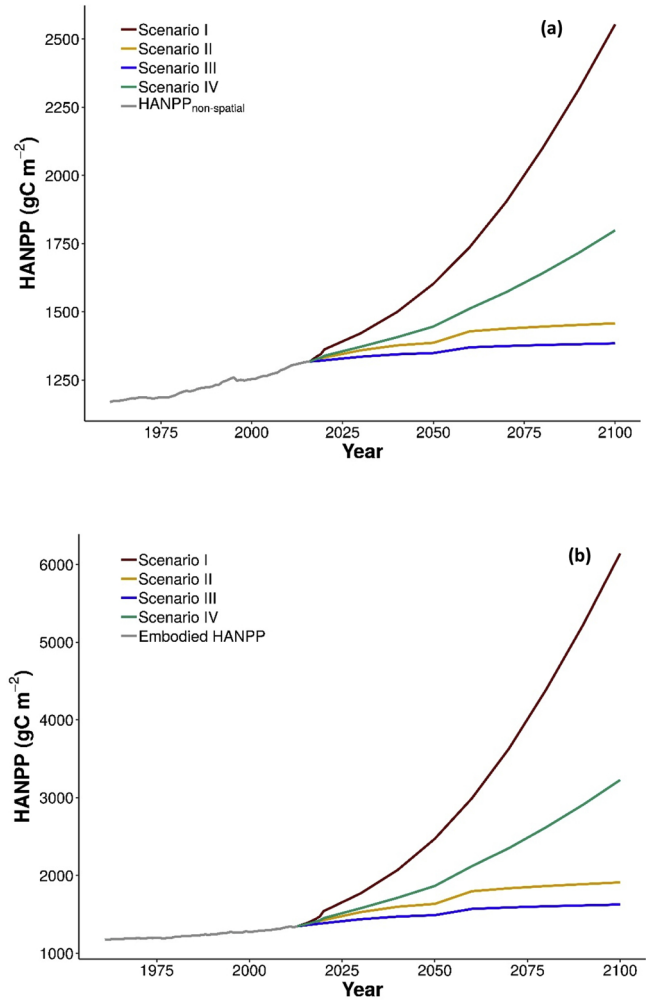


Fig. 5. Future projections of (a) non-spatial HANPP and (b) embodied HANPP using different scenarios.

between biomass consumption and HANPP. Scenario IV encompasses the projections made in Scenario I-III independently, in an aggregate manner. The interpretations of scenario based future projection have been summarized in upcoming section.

4. Discussion

4.1. Interpretation of the results

The spatial HANPP maps (Fig. 2) outline the emergence and spread of agriculture across the major waterbodies of the country. Bangladesh is a riverine country with approximately 700 rivers including tributaries (Khan, 1994) and regions of highest HANPP were observed to be close to major waterbodies such as rivers. It is observed that HANPP of regions near major waterbodies have been consistently increasing and will continue the trend in the future (from green in 1700 to yellow in 1800 to red in 2100). Furthermore, historical spatial HANPP (1700–1800) shows that the whole region of Bangladesh was once filled with natural vegetation, especially due to its geographic position in the tropics with sufficient rainfall and fertile land. India's potential GPP map from DeSouza and Malhi (2018) indicates that there was a huge forested area in Bangladesh confirming a dense tropical forest in this region stretching West Bengal and North East India through Bangladesh to Eastern Indian territory.

From this study, it is evident that Bangladesh once had huge patches of forest which has been reduced to remaining skeleton sized small forest patches. For example, the whole Sylhet region was once a continuous forest, connected with the exterior forest patches from neighbouring countries. The sal (*Shorea robusta*) forest patch in the central part of Bangladesh is no exception either and has been shrinking throughout history. In spite of a high overall HANPP value for Bangladesh, the Chittagong Hill Tracts (CHTs) has low HANPP indicating healthy forest systems, which can be traced to practices and approaches of the indigenous people in the conservation of forests and land productivity (Fig. 6).

The spatial HANPP of Bangladesh also shows that the value of HANPP is high in the urban regions and along the railway tracts. Thus, HANPP can be directly linked to urban growth, long range transport systems and anthropogenic causes in general. Productivity has reached the lowest levels in most urban regions of Bangladesh and so urban forestry could be a significant factor for overcoming this mass land degradation regime.

According to statistical tests performed by Haberl et al. (2014) HANPP is indirectly correlated to species loss. Gaston and Spicer (2013) note habitat loss and degradation as one of the four main reasons behind biodiversity loss and species extinction. Bangladesh, being a developing country, has very poor biodiversity inventory. However, available data suggests that the country has 31 extinct faunal species as well as 56 critically endangered species (IUCN, 2015). Spatial HANPP maps show the high rate of increase of HANPP, implying huge forest loss, which in turn points to loss of both species and species habitat since 80% of terrestrial species live in forests (Gibb et al., 2014). It is crucial to note that there is no mention of Bangladesh in the global conservation priority hotspots proposed by Myers et al. (2000) which gives the list of 25 hotspots where endemic species are undergoing unsustainable loss of habitat. This study instigates the importance of Bangladesh as a potential hotspot for unsustainable loss of species habitat.

4.2. Interpreting future projections of HANPP in Bangladesh

The future projections of HANPP, both non-spatial and embodied, based on aforementioned scenarios show increasing trends of HANPP in the future. As mentioned above, future projections of HANPP based on Scenario I (GDP) shows exponential increase of HANPP of Bangladesh in the future. This can be attributed to the explosive economic development of the country in recent times which will most probably hold true for the near future as well. However, it potentially overestimates the future values of HANPP since economic growth can be stunted/affected by various factors. Scenario II based on the population of the country also shows increasing trend of HANPP mainly due to exponential increase of population over the years. With time as Bangladesh transitions from a developing nation to a developed nation, the population growth rate will decrease, consequently decreasing the rate of increase of HANPP whereas GDP will keep on increasing. Thus, future projections based on Scenario I and II can be directly linked and interpreted to each other even though projected independently. Scenario III, based on biomass consumption, albeit showing an increasing trend of HANPP, has lower magnitude of increase compared to Scenario I and II which can be linked to the transition of Bangladesh from an agricultural nation to an industrial one. Scenario IV, which is an aggregate of Scenario I, II and III, provides the most reliable projections albeit being largely influenced by Scenario I.

4.3. Comparing HANPP of Bangladesh in a global context

Country-wise HANPP values from existing studies are compared with the values obtained in this study. The most obvious and alarming information from Fig. 7 shows that Bangladesh has the highest HANPP of all countries, which is many magnitudes larger than HANPP values of other countries. There may be several major reasons behind such stark differences in values for HANPP. Firstly, potential net primary

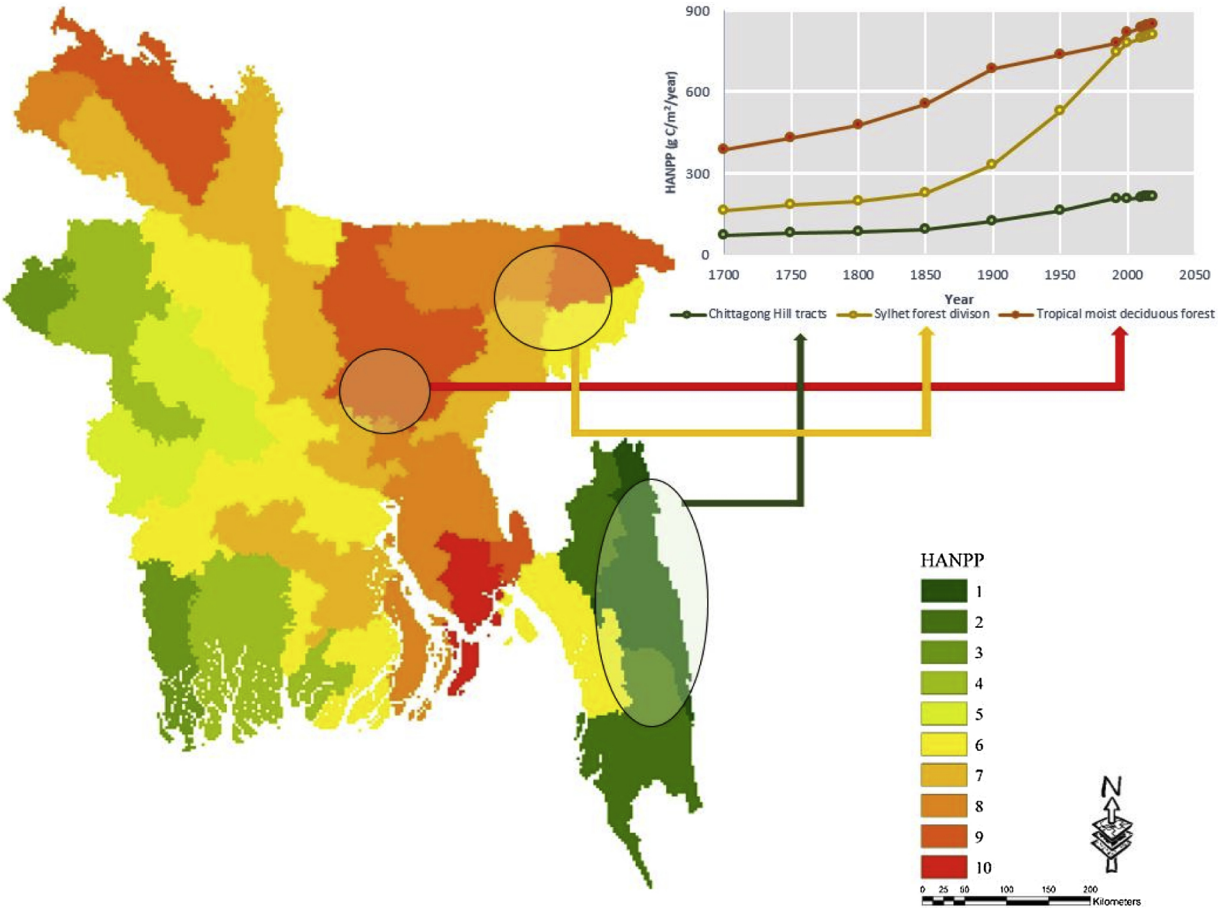


Fig. 6. HANPP in three major forest patches in Bangladesh in 2019.

productivity is higher in Bangladesh than in other countries that are compared in this study. India, holding the second position, has similar geographic and climatic conditions as Bangladesh (most of the European studies followed LPJ DGVM models in order to quantify the potential NPP whereas India and Bangladesh follow a new model developed by DeSouza and Malhi (2018)). However, the mean potential net primary productivity of India and other compared countries are less than that of Bangladesh. Secondly, Bangladesh is the 10th most densely populated country in the world (United Nations, 2019) with a population density of 1278 per km^{-2}. Thirdly, Bangladesh has very poor forest cover area compared to the other countries, which is also evident from the spatial HANPP maps (Fig. 2). Fourthly, Bangladesh has the highest percentage of arable land in the world (World Bank, 2017). European and Scandinavian countries have low HANPP values which can be attributed to factors such as geographic location, forest cover, climatic conditions, socio-economic development approaches etc. Fig. 7b shows that HANPP is correlated to population density of a nation, where Bangladesh has both highest HANPP and highest population density among countries considered. Netherlands is a special case, harbouring both relatively high population density and low HANPP owing to their use of indoor agriculture i.e. hydroponics and aquaponics.

To further validate the findings of this study, we have extracted values for Bangladesh from global studies emphasizing on HANPP (Table 3). While exploring the ratio between HANPP and NPP_0, it is evident that the results of this study is within the range of Imhoff et al. (2004) while being slightly out of range (15.66%) for Haberl et al. (2007) and Haberl et al. (2014). Embodied HANPP is also within range provided in Erb et al. (2009).

5. Implications of the results

Compared to the rapidly growing or developed economies of the world, Bangladesh is the unfortunate champion in HANPP value indicating severe degradation in her natural ecosystem with the country's disproportionate bearing on the ninth planetary boundary. Bangladesh needs to check further increase in HANPP value and take measures to bring it down for sustainable development. HANPP value reported in this paper is the baseline for Bangladesh which the country can use to monitor the sustainability of development interventions by routinely measuring HANPP using reliable and comprehensive databases in reference to this baseline.

As an use case of the estimated HANPP, the distribution of composite HDI-HANPP indicator (Fig. 8) by districts, identifies regions with high HANPP-low HDI (29 districts, Red), high HANPP-high HDI (23 districts, Light Orange), low HANPP-high HDI (7 districts, Grey) and low HANPP-low HDI (5 districts, Green) which provides broad guidelines for designing region specific policies and development plans (see supplementary information for district level results). The low HANPP-high HDI (Green) category included the Sundarbans (the largest chunk of mangrove forest in the world (Iftekhar and Islam, 2004)) and CHTs (excluding Bandarban), mainly due to their forest cover. Evidently, at CHTs, policy and programmes should be tailored to keep shifting cultivation, increasing horticulture, tobacco farming and other forest degradation factors under check while promoting alternative livelihood independent from natural resources, for example ecotourism, to keep HANPP stalled while upscaling forest conservation initiatives to increase NPP. On the other hand, the recent industrial developments in the fringes of the Sundarbans, needs keen monitoring to keep the HANPP at its current state, if cannot be lowered further. Rajshahi is in

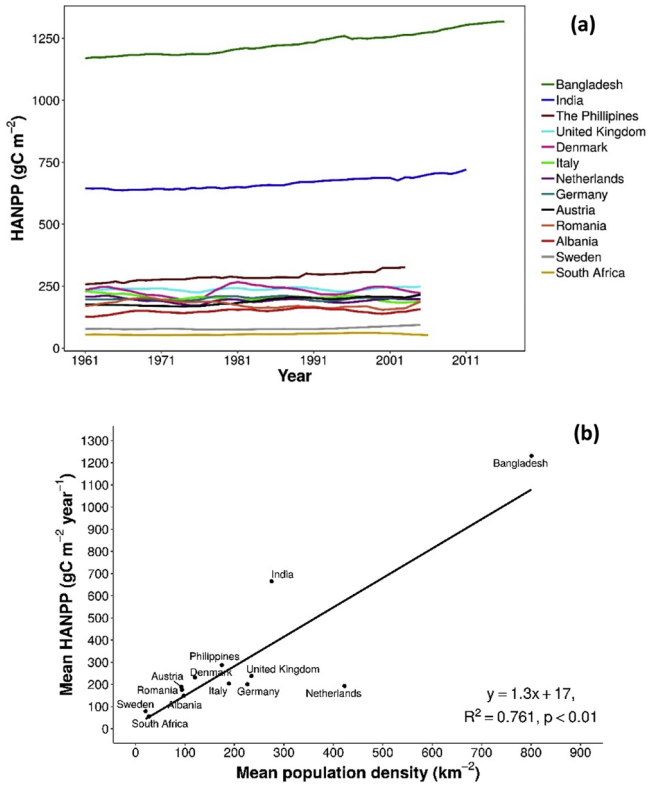


Fig. 7. Global comparison of non-spatial HANPP by country – (a) historical timeline of HANPP and (b) mean HANPP and mean population density (UK, Albania, Sweden, Germany, Italy, Romania, Austria, Netherlands, Denmark: 1961-2005(Gingrich et al., 2015; South Africa: 1961–2006 (Niedertscheider et al., 2012); Spain: 1961–2003 (Schwarzlmüller, 2009); The Phillipines: 1961–2003 (Kastner, 2009); India: 1961–2011 (DeSouza and Malhi, 2018)).

the same zone due to the less primary productivity in the area. In contrary, high HANPP-low HDI in Sylhet region despite having a forest cover is linked to its higher $NPP_{potential}$ than most other regions in Bangladesh. The whole area once covered in forest has experienced a massive land use change due to tea estates, urban and rural sprawls (Hasan et al., 2013; Rai et al., 2017). Hence, enhancement of forest conservation is inevitable for districts falling under high HANPP-low HDI (Red) category, along with taking initiative to bring unused tea-estates in Sylhet under forest cover and acceleration of social forestry activities in the zone. Grey region with low HANPP-low HDI has very high population density and is almost devoid of forests and hence, the study backs social and homestead forestry intervention with extension of affordable, low land-footprint farming technologies to reduce HANPP in this region.

Though higher HANPP value asserts higher rate of land and natural resources degradation (Haberl et al., 2004), there is no definite threshold HANPP value. However, a threshold HANPP may help setting a milestone for development planning specifically with reference to forest and agricultural planning. Accordingly, a gross threshold HANPP is estimated by subtracting NPP_{eco} from $NPP_{potential}$ by assuming NPP of the lowest percentage (25%) forest cover required for ecological balance (MacDicken et al., 2016) as the lowest limit of NPP_{eco}. The value of gross threshold HANPP for Bangladesh becomes 926.675 gC/m^2/year compared to the actual HANPP of Bangladesh. Till a more accurate threshold HANPP is available, the country can orient its policies and regional development planning to increase NPP for bringing down its HANPP to this threshold value. For example, Bangladesh forest policy (Bangladesh Forest Department, 2016) asks to expand the forest cover to 20% of the land area by 2020 which needs to set a target not below 25% for 2030 as indicated by this study. Bangladesh's recent policy and planning (Planning Commission, 2017) seems on track considering the indications of the results of this research though the actual translation of policy wishes, based on historical track, needs to be kept under constant scrutiny.

On the other hand, recommendations to reduce HANPP ask for an estimate of the necessary investments, for which, this research links the GDP (billion 2018 US$/year) and HANPP (kt C/year) for Bangladesh (see supplementary material S6). The result indicates approximately 8.62 million (0.00862 billion) 2018 US$ GDP decrease to lower HANPP by 1 kt C/year. Since forest conservation and afforestation are the main avenues to reduce HANPP, Bangladesh may use this estimate as a guide while seeking investments to lowering its HANPP from financing mechanisms including carbon trading, REDD + etc. Also, donors may use NPP and HANPP as a performance indicator to monitor the outcomes of associated investments.

Quite high HANPP of Bangladesh compared to other countries (Fig. 7) due to its unique geolocation, population density, development trajectory and natural resource management arrested the need to analyse the contribution of each component to the HANPP indicator. It provided insights on the scale of anthropogenic pressure on the regional ecosystems of the country.

The country level estimate of HANPP for Bangladesh will help refine the global HANPP estimates to assist in policy formulation for sustainable natural resource management to attain SDGs in relation to biologically productive land by giving a quantification of the overall anthropogenic pressure on the land system. As the analytical methodology for HANPP estimation is tough to apply in countries lacking good statistical databases - this study is of critical importance regarding the applicability of its approaches and outcomes for many other countries having similar situation in terms of geography, economic transition, demography, development or agronomic pattern as Bangladesh.

Food production and infrastructural development in developing countries, unlike the developed ones, supports the mere survival and inhibits adoption of stringent policies to reduce food production or curb land use changes. In contrary, the need to lower HANPP for sustainable global growth cannot be ignored. The dilemma needs to be addressed by optimizing the land use and food production processes through waste minimization while adopting low land-footprint and ecologically sustainable emerging farming technologies including vertical farming, indoor agriculture, urban farming, aquaponics, floating agriculture etc. In most instances, efforts to reduce HANPP led to the increase in ecological footprint in different countries. In the longer run, though ecological footprints may eventually plateau, policies and technologies should be balanced to make sure that positively tweaking one indicator doesn't upset the other.

Table 3
Comparison of results of this study with global studies showing concurrency.

Quantity	Reference	Year	Referenced value	Results from this study
HANPP/NPP_0	Imhoff et al., 2004	1995	100 to > 200%	116.23%
	Haberl et al., 2007	2000	70 to 100%	115.66%
	Haberl et al., 2014	2000	80 to 100%	115.66%
Embodied HANPP (gC/m^2/year)	Erb et al., 2009	2000	800 to 3000	1275.59

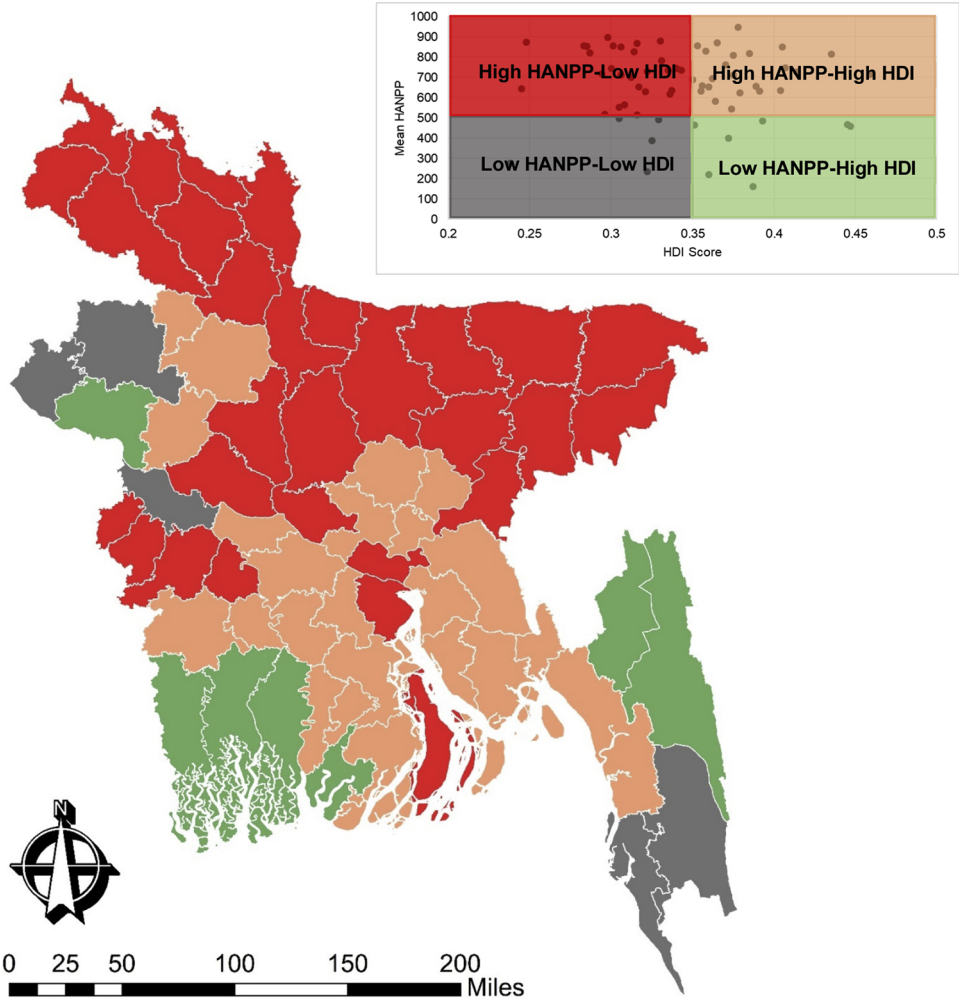


Fig. 8. District wise classification based on HANPP and HDI.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.landusepol.2019.104067.

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